

II. RELATED RESEARCH

A. General Background of OGDCL

OGDCL was established on 1st January 1961 as a public limited company. It is listed on the Pakistan Stock Exchange (PSX) and the London Stock Exchange (LSE). Primarily owned by the Government of Pakistan, with the 85 percent of shares held by the Ministry of Energy (Petroleum Division). However, the company is 15 percent liquid in market. The Company is engaged in exploration, development, and production of oil and gas resources in Pakistan. The company operates in various regions across the country, including on-shore and offshore areas. The company owns and operates numerous oil and gas fields in Pakistan, including some of the largest and most significant fields in the country. These assets contribute significantly to Pakistan's domestic energy production and supply.

B. Fundamental Dependencies of OGDCL

Company's performance is dependent upon two major factors; one is company's production to sales operation pipeline, and second is current financial standing of the company in terms of liquid assets, debt, exchange rate, and SBP policy rate. Then there are country's economic dependencies like circular trade debt is affecting company's revenue collection for smoothly maintaining company's operations.

C. Recurrent Neural Networks

Recurrent Neural Networks (RNNs) are a type of artificial neural network designed to handle sequential data, including multivariate time series data. They are particularly well-suited for tasks where the temporal order of the data is important, such as predicting future values based on historical observations.

D. Contribution of This Paper

This paper focuses on two key areas:

- Dataset preparation of OGDCL company.
- Apply appropriate RNN on prepared dataset such that it produces rational inferences.

III. METHODOLOGY

A. Dataset Generation

Dataset preparation is crucial step where company's historical annual financial statements, income statements, and sales statements are read to prepare a machine readable file format like .xlsx, which can be read by popular machine learning languages like python for loading dataset for doing analysis. These annual reports are available on the company's website [1]. Then SBP historical interest rates are taken from SBP's website, and historical conversion rates are taken from a website named investing.

B. Data Reprocessing

- Due to the limited size of dataset, dataset augmentation would be helpful for increasing size, and quality of dataset.
- As dataset features are at different scales, then dataset normalization is done to mitigate heterogeneous effects of dominating certain features.

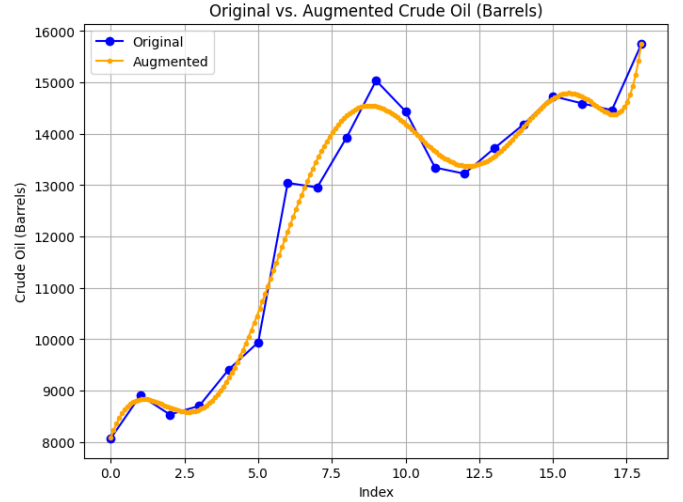


Fig. 2. Oil Production Original vs Augmented

- Dataset is divided into training, validation, and test dataset to do the inference.

C. Model Development

A multi step RNN regression network architecture is devised in order to solve this multivariate regression problem.

- **Input layer:** For taking standardized input.
- **Hidden layers:** LSTM layers for learning relationship.
- **Output layer:** Output layer for regression results.

D. Model Training

Following are the configurations of training

- **Optimizer:** Adam Gradient Descent optimizer.
- **Loss Function:** Measure square error.
- **Call backs:** Implementation of early stopping; to learn best weights and biases.

E. Model Evaluation

Model evaluation for now is for OGDCL only, later on could be tested on other companies like Pakistan Oil Fields Limited, Pakistan Petroleum Limited, or MARI Petroleum Limited. Evaluation would surely be done on real world data to get better, and meaningful insights. I think reasonable evaluation of this model would be test results on multiple companies in order to mitigate effects of anomalies or outlier behavior influenced by management or other external factors. Secondly, model is also going to be evaluated based on general accounting principles, and realistic bound checks.

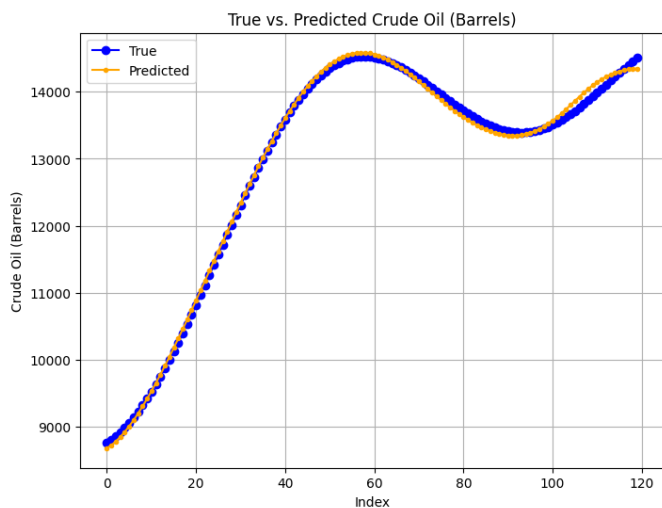


Fig. 3. True vs Predicted oil production on training dataset

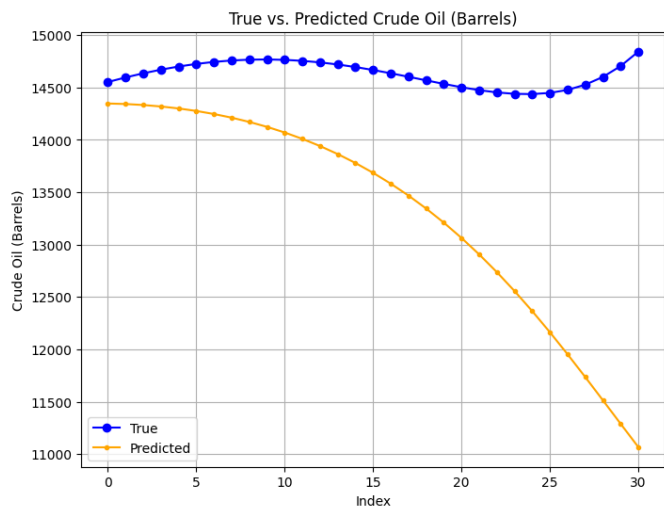


Fig. 6. True vs Predicted sales on testing dataset

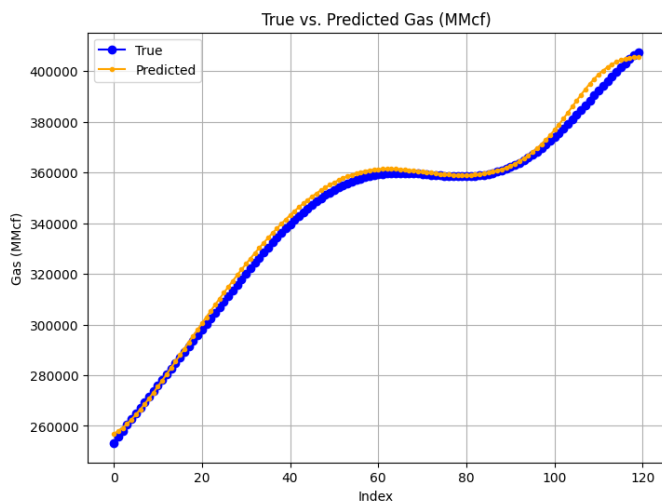


Fig. 4. True vs Predicted gas production on training dataset

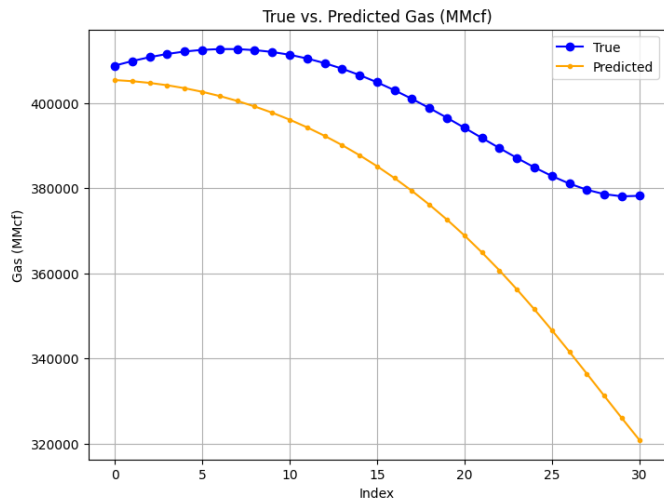


Fig. 7. True vs Predicted sales on testing dataset

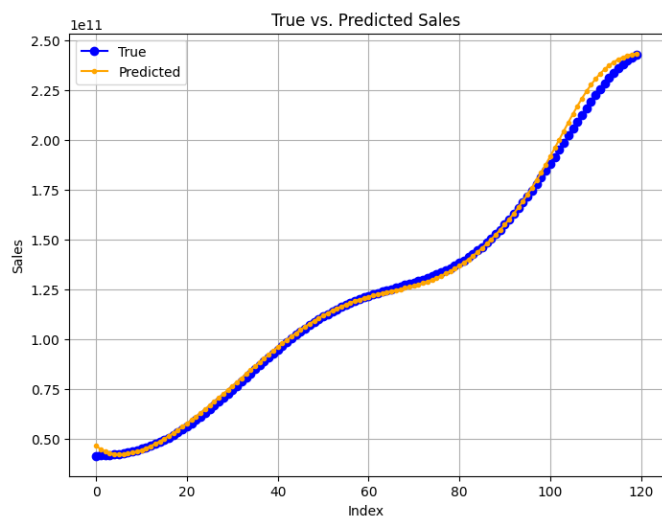


Fig. 5. True vs Predicted sales on training dataset

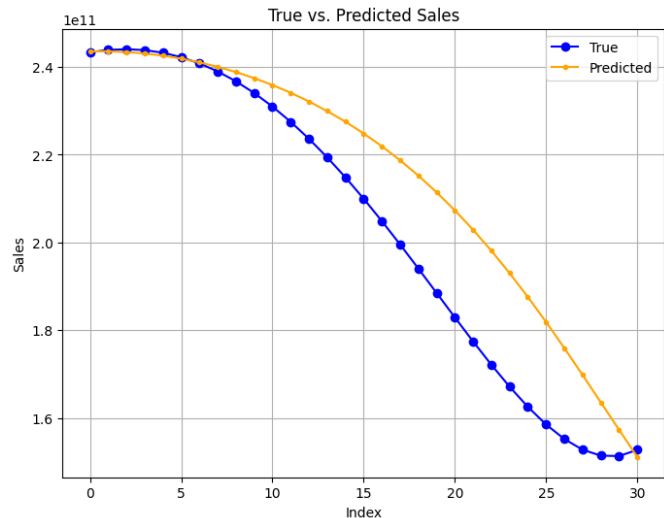


Fig. 8. True vs Predicted sales on testing dataset

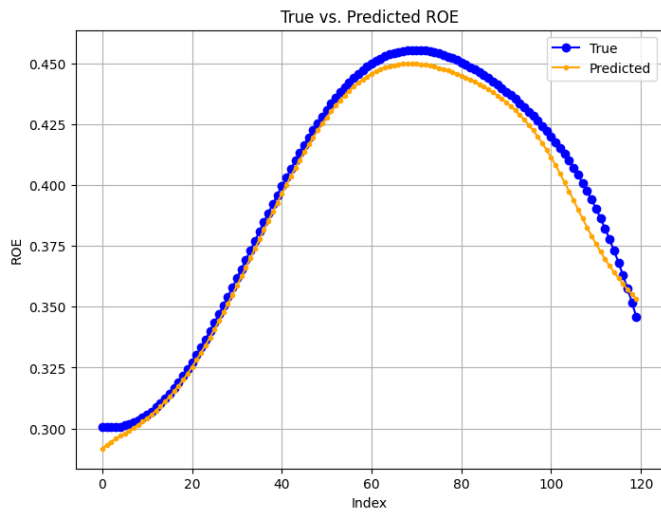


Fig. 9. True vs Predicted ROE on training dataset

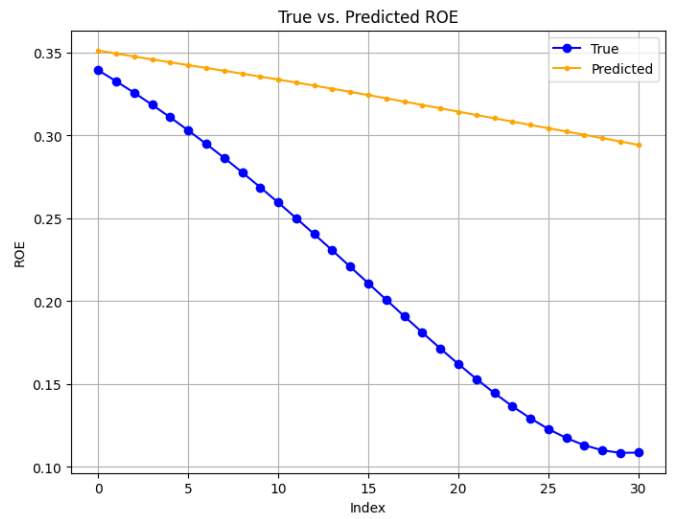


Fig. 12. True vs Predicted ROE on testing dataset

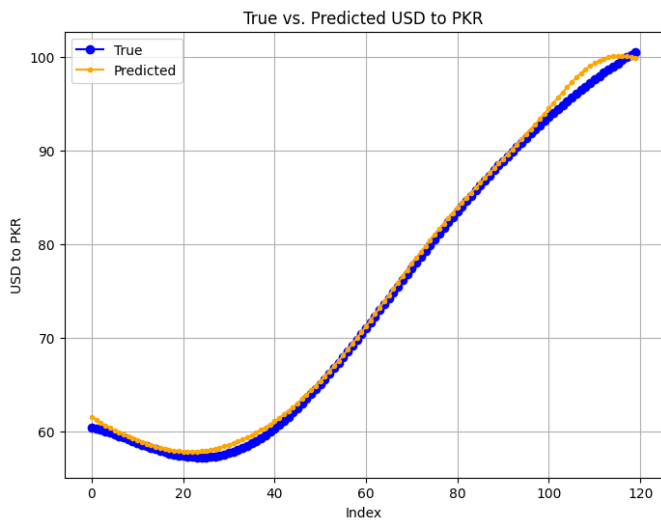


Fig. 10. True vs Predicted USD exchange on training dataset

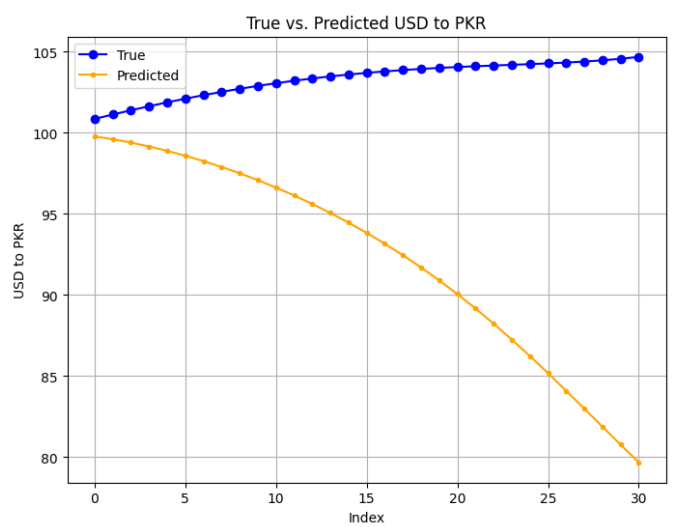


Fig. 13. True vs Predicted USD to PKR exchange on testing dataset

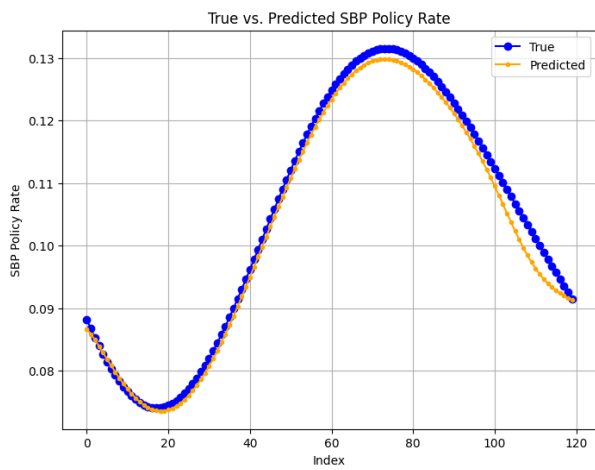


Fig. 11. True vs Predicted policy rate training

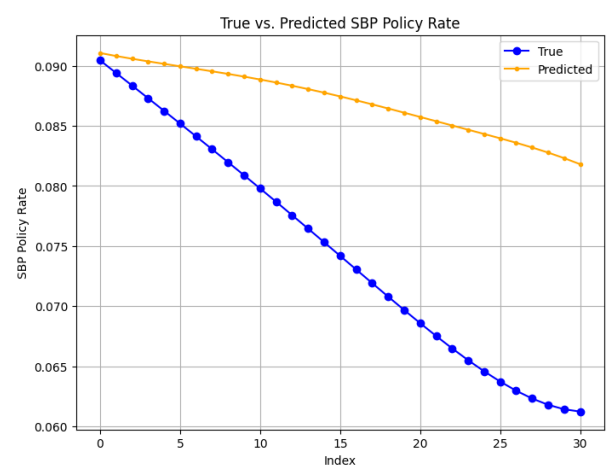


Fig. 14. True vs Predicted SBP policy rate on testing dataset

F. Commentating on Results

Results are in rational bounds, but due to unexpected change in country's economy like dollar parity, policy rate hikes, and variations in drilling and discoveries that are not accounted impacted results in the form of deviation from true results.

IV. RESULTS

Model is evaluated based on training, validation, and test dataset, and results were shown in from figure 3 till 14.

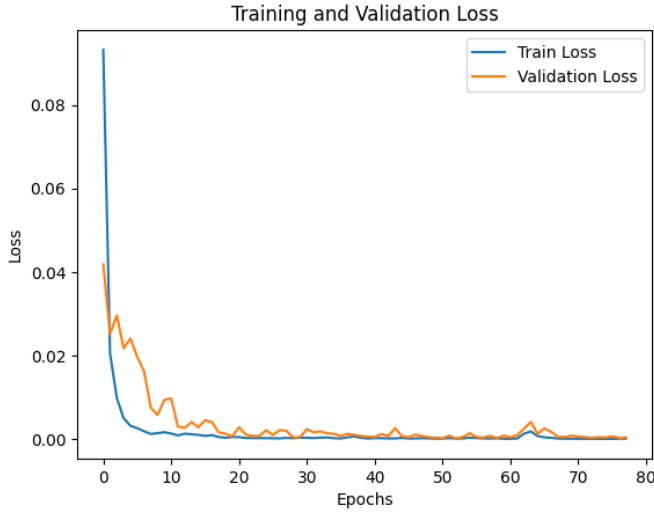


Fig. 15. Training and Validation Loss Evaluation

Despite training and validation gap is minimized, but inferences are not rational at this stage by applying basic accounting principles.

V. CONCLUSION

Nothing conclusive can be deduced at this stage as dataset preprocessing revision is proposed after meeting with Dr. Tahira, and model needs revision as well.

ACKNOWLEDGMENT

I would like to thank Dr. Tahira for encouraging us to learn LaTeX and write papers throughout the Machine Learning course journey. Additionally, I would like to express gratitude to our teaching assistant, Miss Malaika Waheed, for her dedicated efforts in assisting us throughout the course.

REFERENCES

- [1] Siraj Khan, "Prepared Dataset from OGDCL's Financial Statements," Accessed from OGDCL's website: <https://www.ogdcl.com/all-financial-reports>.
- [2] Historical SBP interest rates, Accessed from: <https://www.sbp.org.pk/ecodata/OVR-Repo-History.pdf>.
- [3] Historical exchange rates, Down sampled and averaged after accessing from: <https://www.investing.com/currencies/usd-pkr-historical-data>.